



M7XC Compact Satellite Modem Manual

Appendix F: Point to Multi-Point VCM / ACM Operation



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NOTE: All appendices may not be present in manual. Some appendixes may be shipped with the option.

1 **Preface**

This Appendix is to be used in conjunction with the main M7XC manual. The manual and appendices are intended for use by personnel with reasonable knowledge of satellite communications and earth station terminology. This manual provides Installation, Operating, and Maintenance procedures for the M7XC Compact Satellite Modem and available options.

Revision History

Release Date	Comments
06/16/2024	Initial Public Release

2 Point to Multi-Point VCM / ACM Operation Overview and Theory

In a Point to Multi-Point network environment a common operational configuration is to configure the “outbound traffic” (from Hub site to Remote Sites) as a single broadcast carrier from hub to satellite and satellite to remote. Traffic that is destined to a particular site or broadcast and multi-cast traffic destined to multiple sites are decoded and ingested by the appropriate site(s).

Returning traffic (often called inbound traffic) is generated at the remote site location and sent back to the Hub. This return traffic can be via FDMA SCPC carrier, Dynamic Carrier Assignment or co-exist on shared carriers in the time domain such as TDMA. For the purposes of this Appendix, we will focus on FDMA return carriers.

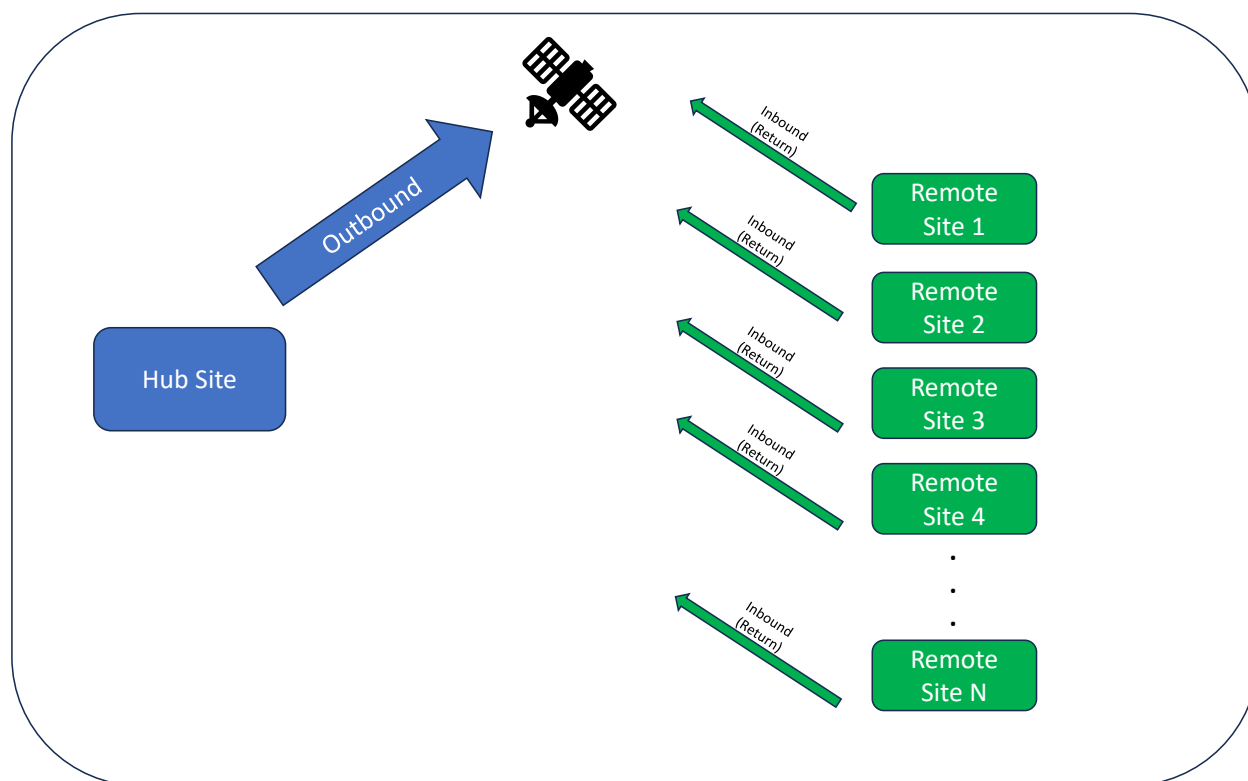


Figure 2-1 VCM/ACM Overview

3 Outbound Traffic Optimization

In a shared outbound Pt to Multi-Pt configuration as shown in Figure 2-1, even though all sites are receiving and demodulating the same signal from the Hub location, differences in antenna size, weather conditions, position within the satellite beam or mobility applications can show large variations in receive signal quality and signal strength.

An outbound Hub carrier that is received by multiple remote sites at varying quality levels historically meant that all of the outbound user traffic had to be “protected” with a single, fixed modulation and coding scheme (MODCOD) that ensured the most disadvantaged site could demodulate and decode the signal as shown in the below example (Figure 3-1).

Site 1 has the lowest perceived signal quality due to being a mobile site with a small antenna. The demodulator Es/No reading of 2.5dB is the weakest in the example network. Historical networks would limit all of the Hub Outbound traffic to a MODCOD that could be locked to signal of 2.5dB or lower without errors to guarantee link availability to all sites.

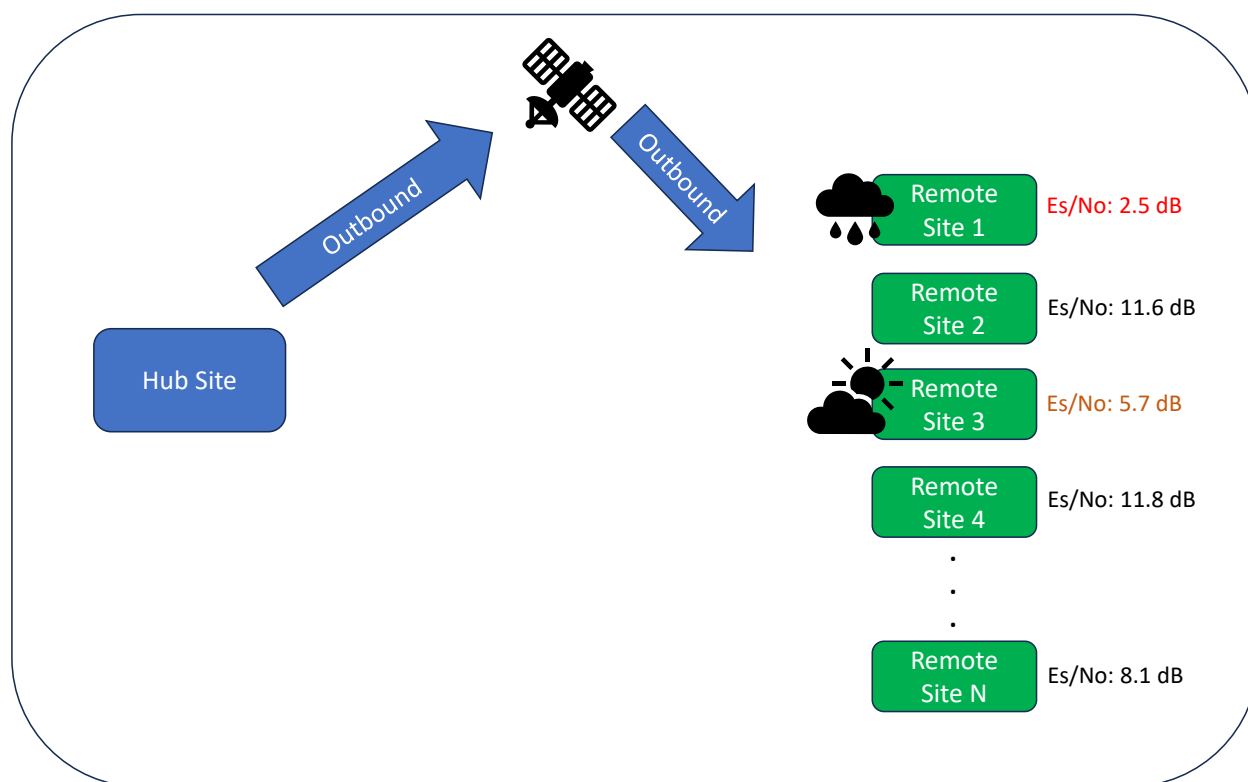


Figure 3-1 VCM/ACM with Various Remote Es/No

A more modern method to deal with varied remote site signal quality is to compartmentalize the Outbound user traffic into bursts that are encapsulated with varying levels of protection. The method is commonly referred to as ACM (Adaptive Coding and Modulation). In this way the outbound aggregate traffic is packaged with multiple levels of protection such that sites that benefit from a higher signal quality can receive higher MODCODs when their signal is strong and maximize user traffic while other sites will use more robust signal protection at the expense of lower user throughput for the same BW used. Datum offers a dynamic outbound ACM operation where each site is dynamically sent data in the best fit MODCOD rather than a fixed MODCOD for the site or the network.

Using the remote site demodulator Es/No values and looking at the DVB-S2X MODCOD table, we can see the best fit MODCOD per site to be:

DVB-S2X Normal Block				VLAN MAPPING
Performance				
MOD	FEC	Spec Eff (Bits / Hz)	QEF Es/No	
QPSK	1/4	0.4800	-2.35	
QPSK	13/45	0.5543	-2.03	
QPSK	1/3	0.6400	-1.24	
QPSK	2/5	0.7700	-0.30	
QPSK	9/20	0.8680	0.22	
QPSK	1/2	0.9653	1.00	
QPSK	11/20	1.0627	1.45	
QPSK	3/5	1.1600	2.23	Site 1
QPSK	2/3	1.2908	3.10	
QPSK	3/4	1.4521	4.03	
QPSK	4/5	1.5494	4.68	
8PSK	3/5	1.7400	5.50	Site 3
8PSK	23/36	1.8531	6.12	
16APSK	26/45	2.2275	7.51	
16APSK	3/5	2.3138	7.80	
16APSK	28/45	2.4001	8.10	
16APSK	23/36	2.4648	8.38	
16APSK	2/3	2.5750	8.97	
16APSK	25/36	2.6806	9.27	
16APSK	13/18	2.7884	9.71	
16APSK	3/4	2.8960	10.21	
16APSK	7/9	3.0042	10.65	
16APSK	4/5	3.0900	11.03	Site 2
16APSK	5/6	3.2220	11.61	
16APSK	77/90	3.3062	11.99	
32APSK	32/45	3.4344	11.75	Site 4
32APSK	11/15	3.5424	12.17	
32APSK	3/4	3.6230	12.73	

Figure 3-2 Best Fit MODCOD By Site #

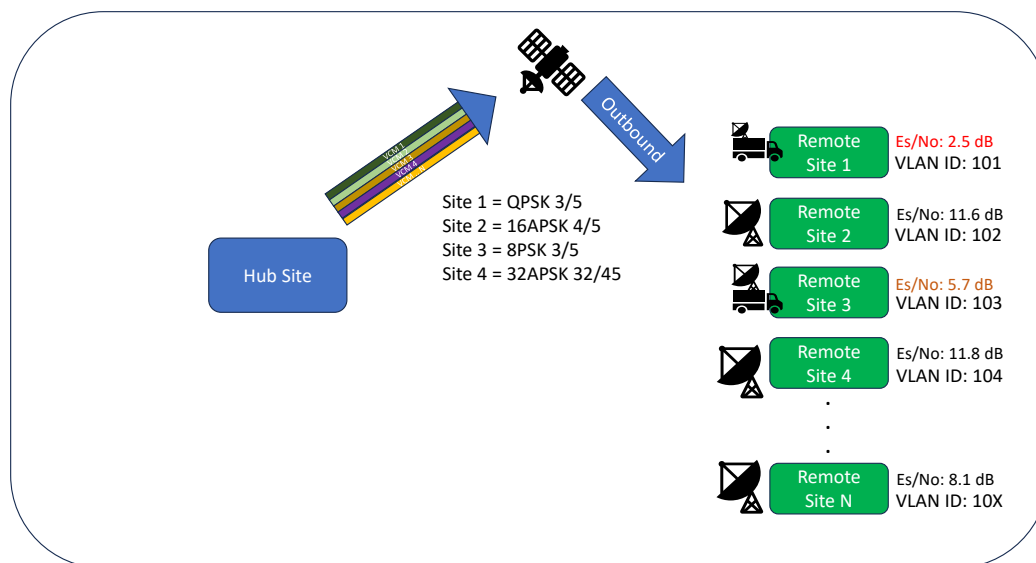


Figure 3-3 Site Best Fit MODCOD

In the ACM environment a shared Outbound carrier contains one or more MODCODs that in aggregate, make up the full Outbound carrier. A “Paired Rx Channel” represents a unique “virtual” Hub to remote carrier that has it’s own specific best-fit MODCOD for that remote at that specific time. A “Paired Rx Channel” can have one or more Queues associated with that virtual carrier. Each Queue is associated with a VLAN tag as seen by the Modulator’s Ingress Buffer. See below for how this is mapped:

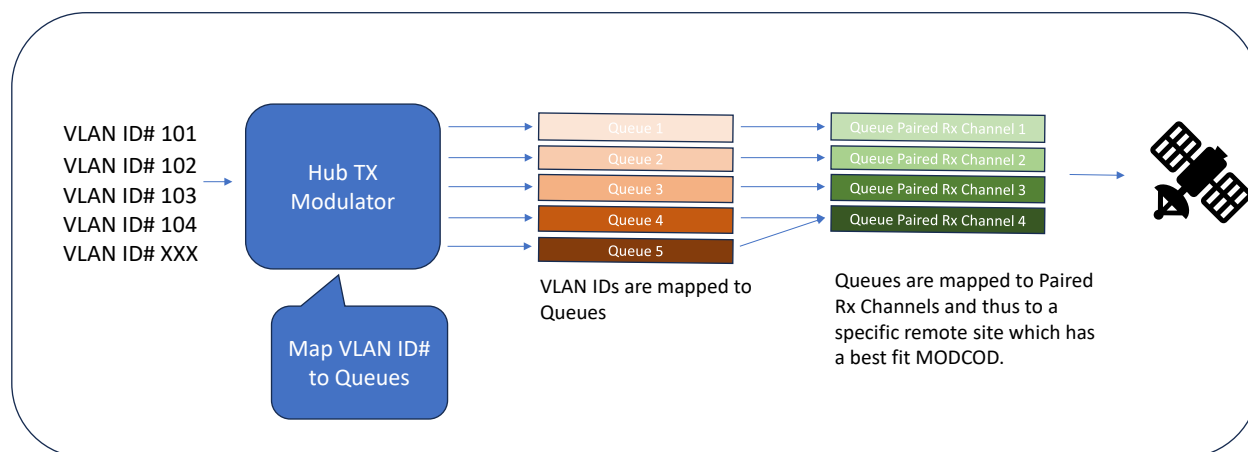


Figure 3-4 Mapping VLAN Tag > Queue > Paired Rx Carrier

The Hub Outbound traffic will now be an aggregate of all sites traffic using the best fit MODCOD for those sites

4 Configuring VCM/ACM Operation on M7XC modems

Now that we understand the theory of Outbound VCM operation and the benefits of maximizing the satellite efficiency on a per site basis we can configure the modems to function in a VCM environment.

For the purposes of this manual, we will associate the following names to the various modems in the VCM/ACM Network:

Hub TX/RX Modem: The M7XC configured as the Tx modem at the Hub site that is responsible for transmitting the shared Tx outbound carrier and multi-demod landing up to 16 sites (Flex LDPC) or 2 sites (DVB-S2X).

Hub RX-Only Demod: One or more M7XC modems configured as a demod or multi-demod ONLY. This Rx-Only demod is capable of landing up to sixteen (16) Flex LDPC sites, or two (2) DVB-S2X sites, or one (1) site running DSSS. Note: Hub Rx-Only Demods are ONLY needed if the VCM/ACM network requires more sites that the Hub RX/RX Modem is capable of. The TOTAL number of sites that the VCM/ACM network can have can not exceed 16 sites.

Remote Modem: A remote site in a VCM/ACM network

4.1 VCM/ACM Network Modem Count Examples:

Network 1: High Density 16 Sites (all remote return carriers are within 100MHz continuous bandwidth)

Hub Site	Waveforms	Count
Hub TX/RX Modem	TX: DVB-S2X VCM RX: (16) Flex LDPC	1
Remote Sites:	Waveforms	Count
Remote Modem	RX: DVB-S2X VCM TX: Flex LDPC	16

Network 2: High Capacity 16 Sites (each set of 2 DVB-S2X return carriers are within 100MHz continuous bandwidth per set)

Hub Site	Waveforms	Count
Hub TX/RX Modem	TX: DVB-S2X VCM RX: (2) DVB-S2X	1
Hub RX-Only Demod	TX: None RX: (2) DVB-S2X	7

Remote Sites:	Waveforms	Count
Remote Modem	RX: DVB-S2X VCM TX: DVB-S2X	16

Network 3: Mixed Use 16 Sites (8 LDPC carriers, 6 High Capacity DVB-S2X return carriers and 2 DSSS return carriers)

Hub Site	Waveforms	Count
Hub TX/RX Modem	TX: DVB-S2X VCM RX: (8) Flex LDPC	1
Hub RX-Only Demod	TX: None RX: (2) DVB-S2X	3
Hub RX-Only Demod	TX: None RX: (1) DSSS	2
Remote Sites:	Waveforms	Count
Remote Modem	RX: DVB-S2X VCM TX: Flex LDPC	8
Remote Modem	RX: DVB-S2X VCM TX: DVB-S2X	6
Remote Modem	RX: DVB-S2X VCM TX: DSSS	2

4.2 Outbound VCM/ACM VLAN Tags

VCM works in conjunction with VLAN 802.1Q tagging. Ideally, user traffic at the Hub site that is destined for the remote sites should be **pre-identified and pre-tagged by an external router / switch or 3rd party device**. Associating a subnet or set of IP / MAC address to a VLAN ID of a remote site is outside of the scope of this Appendix and should be discussed with the proper networking staff. Using one mechanism or another, any traffic that is strictly destined for a specific remote site should be encapsulated with the outermost VLAN tag unique to that remote site. VLAN tags for broadcast or multicast traffic can also be configured by the networking staff.

Once the Hub to Remote user traffic is properly tagged the hub modem can be configured to perform a lookup to associate a VLAN tag or VLAN range to a VCM selection and therefore a MODCOD level of protection.

5 VCM ACM Network Example

The following sections to configure an actual VCM/ACM network will use the below diagrams as a reference for the overall network configuration. It is suggested to create a similar diagram for any real network planning purposes.

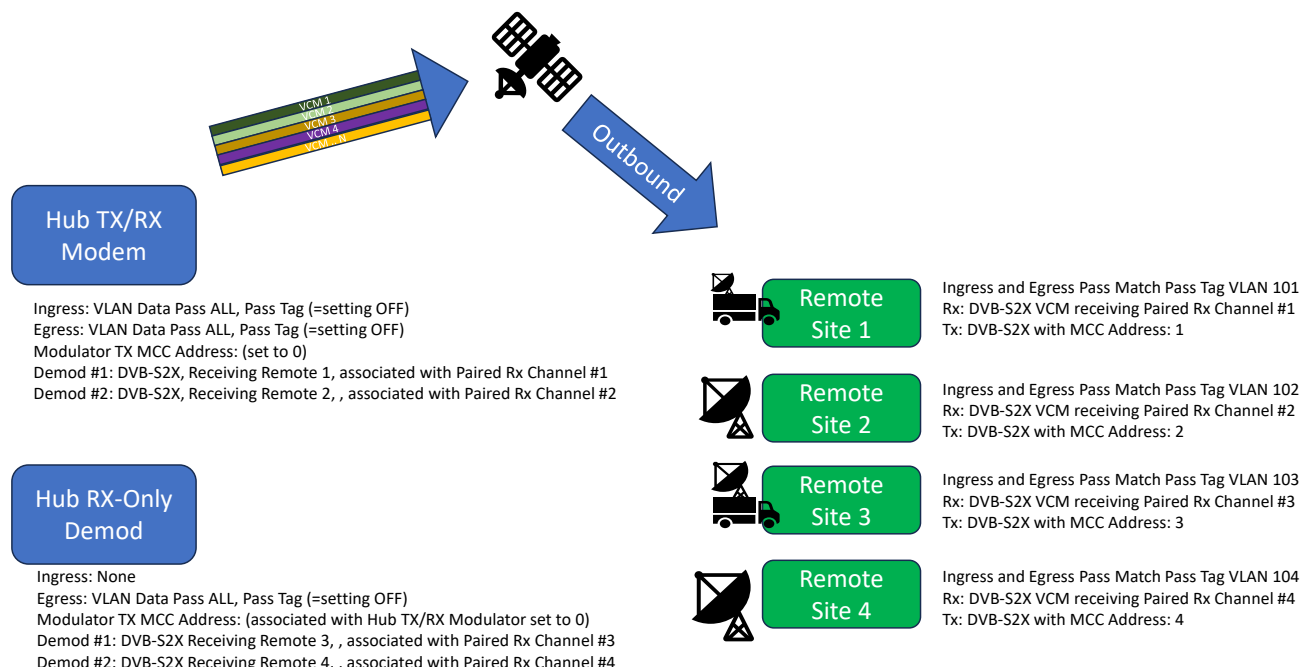


Figure 5-1 Sample Network

6 Hub TX/RX Modem Configuration

6.1 Hub TX/RX Modem >Unit>WAN Data Port Configuration

The Hub modulator is responsible for accepting all user traffic from the LAN side of the modulator and moving the traffic into the proper VCM bucket. To do this, it must accept all traffic or optionally accept a “range” of traffic that is associated with the VLAN ID’s of the remote sites or configured broadcast VLAN IDs. Goto the Unit>WAN Data Port page of the Hub Modem GUI. Configure the Hub Modulator’s Unit/WAN Data Port VLAN filters in the following way:

Table 5-1 VLAN Ingress Tag Filter Selections

Ingress VLAN Tag Filter	Operation
Off	Will accept all user traffic with or without VLAN tags and pass them to the WAN for transmission
Pass Match, Strip Tag	Will only accept VLAN tags within a user defined range into the modem. It will associate the packet with the proper VCM bucket / queue but strip the VLAN tag before sending to the satellite WAN. In this way the remote will NOT see a VLAN tag

Pass, Append Tag	Will accept all user traffic with or without VLAN tags and will “PUSH” a new user defined VLAN tag on all packets. NOTE: Not suggested at Hub Modulator
Pass Match, Pass Tag	Modulator will only pass those packets with matching VLAN tag, and will keep the VLAN tag on the packet when sent over the satellite WAN. NOTE: Not suggested at Hub Modulator
Pass Range Match, Pass Tag	Modulator will only pass those packets with a VLAN tag within the user defined range of VLAN IDs and will keep the VLAN tag on the packet when sent over the satellite WAN.

The most common configurations of the Hub Modulator Ingress VLAN Tag Filter are:

1. Off - send all traffic
2. Pass Range Match, Pass Tag – Only send traffic that is within the VLANs I care about
3. Pass Match, Strip Tag – Used when the remotes are not going to filter on VLANs

Similarly, a selection is made for the Hub Modulator’s Egress VLAN Tag Filter or what exits the modems LAN port:

Table 5-2 VLAN Egress Tag Filter Selections

Egress VLAN Tag Filter	Operation
Off	Will accept all user traffic with or without VLAN tags and pass them to the WAN for transmission
Pass, Append Tag	Will accept all traffic and PUSH a user defined VLAN ID tag onto each packet
Pass Match, Strip Tag	Modulator will only pass those packets with matching VLAN tag, and will remove the VLAN tag on the packet when sent onto the LAN
Pass Match, Pass Tag	Modulator will only pass those packets with matching VLAN tag, and will keep the VLAN tag on the packet when sent onto the LAN
Pass Range Match, Pass Tag	Modulator will only pass those packets with a VLAN tag within the user defined range of VLAN IDs and will keep the VLAN tag on the packet when sent onto the LAN.

Selections are made on the GUI see Figure 5-1

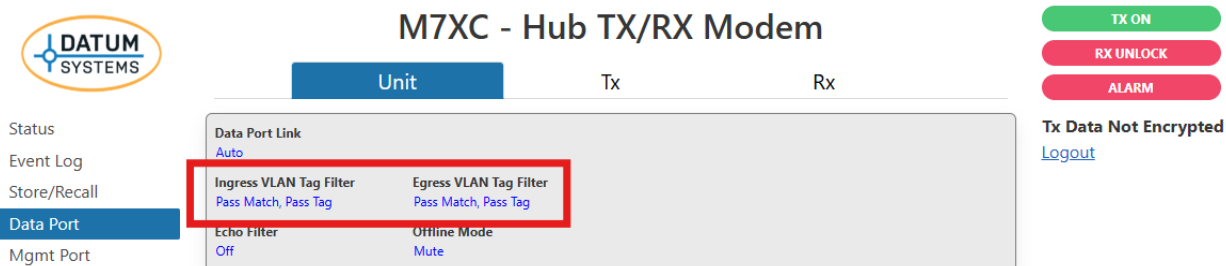


Figure 6-1 HUB TX/RX Modem >Unit>WAN GUI

The Remote modems will also be configured in a similar manner to either accept all traffic, a specific VLAN ID or a range of VLAN IDs and pass or POP the actual tag.

6.2 Hub TX/RX Modem >Tx>Modulator VCM/ACM Configuration

Configure the Hub to Remote or “Outbound” carrier. This is the outbound carrier that all remotes will demodulate and will decode and ingest data that is specifically destined for a particular remote or set of remotes from the Hub. This outbound carrier configuration will have many common properties for all remotes such as the carrier frequency, symbol rate and ROF values. Data within this common carrier is segregated by VLAN and placed into one or more “Queues”.

The common settings of the outbound carrier are all marked with an asterisk “*”. The common settings are found in the upper section of the >Tx>Modulator tab when the waveform is set to “DVB-S2X VCM”.

M7XC - Hub TX/RX Modem

Unit **Tx** Rx

Waveform * **DVB-S2X VCM**

Queue ModCod QPSK 1/4 Normal with Pilot, 23.93 Mbps, QEF @ -2.35 dB

Output * **On** Spectrum * **Normal** Test Modulation * **Off** Mute * **Automatic**

Carrier Frequency * **1.000000000 GHz** Tilt * **0.00 dB** Roll-Off * **10%**

Level * **-2.00 dBm**

Symbol Rate * **50.000000 Msp/s** Frame Size * **Normal** Pilot * **On** PLS Scrambler Sequence * **0**

Queue Service **DVB-S2X ACM Preferred** Queue VLAN ID **101** Queue Paired Rx Channel **1** Queue CIR Mode **Normal**

Queue Min Es/N0 **-5.00 dB** Queue Max Es/N0 **10.00 dB** Desired Queue CIR **1.000000 Mbps** Queue MIR **600.000000 Mbps**

Desired Drain Time * **100 ms** Drain Time @ Min ModCods * **2.66 ms** Total Desired CIR * **91.103680 Mbps** Total CIR @ Min ModCods * **23.928850 Mbps**

Queue Live Time * **500 ms** Queue Remote ID **Remote 1**

ACM Margin * **0.50 dB** ACM 4-ary Hysteresis * **0.00 dB** ACM 8-ary Hysteresis * **0.10 dB** ACM 16-ary Hysteresis * **0.10 dB**

ACM 32-ary Hysteresis * **0.10 dB** ACM 64-ary Hysteresis * **0.20 dB** ACM 128-ary Hysteresis * **0.20 dB** ACM 256-ary Hysteresis * **0.30 dB**

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TX ON RX UNLOCK ALARM

Queue 1 **CIR MAX ALM**

Queue 2 **CIR MAX ALM**

Queue 3 **CIR MAX ALM**

Queue 4 **CIR MAX ALM**

Queue 5 **Off**

Queue 6 **Off**

Queue 7 **Off**

Queue 8 **Off**

Queue 9 **Off**

Queue 10 **Off**

Queue 11 **Off**

Queue 12 **Off**

Queue 13 **Off**

Queue 14 **Off**

Queue 15 **Off**

Queue 16 **Off**

* Common To All Queues
Tx Data Not Encrypted
[Logout](#)

Figure 6-2 Hub TX/RX Modulator >Tx>Modulator TX Carrier Common Settings

- **Waveform*:** The waveform should be must be set to “DVB-S2X VCM” to allow Pt-Multi-Pt or Hub/Spoke operation where the outbound carrier will be running VCM or ACM. A setting of “DVB-S2X” or any other waveform can be used for Pt-Multi-Pt operation but the system will ONLY be capable of CCM operation in that configuration. DVB-S2X VCM is nominal setting.
- **Carrier Frequency*:** Set the DVB-S2X outbound carrier center frequency. This is the frequency all remotes demods will be tuned to.
- **Symbol Rate*:** Set the DVB-S2X outbound symbol rate. This is the symbol rate all remotes demods will be tuned to.
- **Frame Size*, Pilot*, PLS Scrambler*:** Set these parameters according to normal carrier parameters for the outbound carrier. These parameters must match the remote demods parameters for all settings. Normal settings user are:
 - Frame Size: Normal
 - Pilot: On
 - PLS Scrambler: 0

M7XC - Hub TX/RX Modem

Unit **Tx** Rx

Waveform *
DVB-S2X VCM

Queue ModCod
QPSK 1/4 Normal with Pilot, 23.93 Mbps, QEF @ -2.35 dB

Output * **Spectrum *** **Test Modulation *** **Mute ***
On Normal Off Automatic

Carrier Frequency * **Tilt *** **Roll-Off ***
1.000000000 GHz 0.00 dB 10%

Level *
-2.00 dBm

Symbol Rate * **Frame Size *** **Pilot *** **PLS Scrambler Sequence ***
50.0000000 Msps Normal On 0

Queue Service **Queue VLAN ID** **Queue Paired Rx Channel** **Queue CIR Mode**
DVB-S2X ACM Preferred 101 1 Normal

Queue Min Es/N0 **Queue Max Es/N0** **Desired Queue CIR** **Queue MIR**
-5.00 dB 10.00 dB 1.000000 Mbps 600.000000 Mbps

Desired Drain Time * **Drain Time @ Min ModCods *** **Total Desired CIR *** **Total CIR @ Min ModCods ***
100 ms 2.66 ms 91.103680 Mbps 23.928850 Mbps

Queue Live Time * **Queue Remote ID**
500 ms Remote 1

ACM Margin * **ACM 4-ary Hysteresis *** **ACM 8-ary Hysteresis *** **ACM 16-ary Hysteresis ***
0.50 dB 0.00 dB 0.10 dB 0.10 dB

ACM 32-ary Hysteresis * **ACM 64-ary Hysteresis *** **ACM 128-ary Hysteresis *** **ACM 256-ary Hysteresis ***
0.10 dB 0.20 dB 0.20 dB 0.30 dB

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Queue 1
CIR MAX ALM

Queue 2 CIR MAX ALM

Queue 3 CIR MAX ALM

Queue 4 CIR MAX ALM

Queue 5 Off

Queue 6 Off

Queue 7 Off

Queue 8 Off

Queue 9 Off

Queue 10 Off

Queue 11 Off

Queue 12 Off

Queue 13 Off

Queue 14 Off

Queue 15 Off

Queue 16 Off

* Common To All Queues
Tx Data Not Encrypted

Figure 6-3 Hub TX/RX Modulator >Tx>Modulator Queue Specific Settings

- Queue Service: This can be set Per Queue (per Remote site)
 - Off: This Queue will not be used
 - DVB-S2X (ACM to the site is not enabled)
 - DVB-S2X ACM Preferred: Typical MODCODs used in a DVB-S2X outbound with ACM. Use this setting for most applications where the linearity of the outbound carrier is typical or not known.
 - DVB-S2X ACM Preferred-L: Use ONLY if your Hub outbound carrier is extremely linear (this will give you a small user throughput advantage over DVB-S2X ACM Preferred but only if the carrier is nearly ideal and has near linear amplifier backoff)
 - Broadcast to All Destinations: Can ONLY be used when configuring Queue 1. This queue and Queue Service can be used if broadcast (destined to all remotes) has been tagged with a specific VLAN ID prior to hitting the Hub TX/RX modem. All traffic filling this Queue will be sent to all remotes at the lowest common MODCOD used by any one remote.
- Queue VLAN ID: This is the outermost VLAN tag of a VLAN tagged Ethernet frame. If the outmost VLAN tag matches the Queue VLAN ID that Ethernet frame will be sent to the corresponding Queue being currently configured (in this example Queue 3)

- Queue Paired RX Channel: This is the channel identifier that will be embedded in the sent message, a remote site with the matching Queue Paired RX Channel identifier will be allowed to demodulate and decode this user traffic.
- Queue CIR Mode: Normal is the nominal setting. A setting of Periodic will allow other queues to drain out of order in rare cases of low latency settings.
- Queue Min Es/No: User selected between -5.0 and 1.99dB. This setting will essentially give a floor to the lowest MODCOD to be used for the Queue in cases of poor weather or link outage. The lower the value allowed the more robust the link will stay up but the lower the user throughput. A user may want to set this the minimum to allow the link to stay up as long as possible or may want to set it to a higher level such as 1 or 2 so that VERY low MODCODs are not used to try and attain the "Desired Queue CIR". A low MODCOD combined with an excessive Desired CIR can result in too many frames being used to attain one Queue's CIR and thus penalizing the entire network.
- Queue Max Es/No: User selected between -4.99 and 25dB. This setting will essentially give a ceiling to the maximum MODCOD to be used for the Queue. A ceiling MODCOD may be desirable to limit the user traffic to a particular site or to prevent higher order MODCODs such as 64APSK and > from running if the link is not highly linear.
- Desired Queue CIR: This is the minimum user Datarate (committed information rate) that is desired for use by the currently selected Queue. The Hub modulator will attempt to give more or less outbound DVB-S2X frames to satisfy this Queue's Desired CIR. A low MODCOD setting or a low Queue Min Es/No may require a substantial percentage of the hub TX carrier to meet high Desired CIR figures when running at low MODCODs.
- ***Repeat all steps above to configure the remaining Queues in the network.***

M7XC - Hub 1

Unit: Tx Rx

Modulator Queue Common Settings

Waveform *	Queue ModCod		
DVB-S2X VCM	16APSK 25/36 Normal with Pilot, 134.03 Mbps, QEF @ 9.27 dB		
Output *	Spectrum *	Test Modulation *	Mute *
On	Normal	Off	Automatic
Carrier Frequency *	Tilt *	Roll-Off *	
1.000000000 GHz	0.00 dB	10%	
Level *			
-2.00 dBm			
Symbol Rate *	Frame Size *	Pilot *	PLS Scrambler Sequence *
50.0000000 Msps	Normal	On	0
Queue Service	Queue VLAN ID	Queue Paired Rx Channel	Queue CIR Mode
DVB-S2X ACM Preferred	103	3	Normal
Queue Min Es/N0	Queue Max Es/N0	Desired Queue CIR	
-5.00 dB	10.00 dB	1.0000000 Mbps	
Desired Drain Time *	Drain Time @ Min ModCods *	Total Desired CIR *	Total CIR @ Min ModCods *
100 ms	2.66 ms	4.107360 Mbps	23.928850 Mbps
Queue Live Time *	Queue Remote ID		
500 ms	Remote 3		
ACM Margin *	ACM 4-ary Hysteresis *	ACM 8-ary Hysteresis *	ACM 16-ary Hysteresis *
0.50 dB	0.00 dB	0.10 dB	0.10 dB
ACM 32-ary Hysteresis *	ACM 64-ary Hysteresis *	ACM 128-ary Hysteresis *	ACM 256-ary Hysteresis *
0.10 dB	0.20 dB	0.20 dB	0.30 dB

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Queue 1: ACM MIN ALM
Queue 2: ACM
Queue 3: ACM
Queue 4: ACM
Queue 5: Off
Queue 6: Off
Queue 7: Off
Queue 8: Off
Queue 9: Off
Queue 10: Off
Queue 11: Off
Queue 12: Off
Queue 13: Off
Queue 14: Off
Queue 15: Off
Queue 16: Off

* Common To All Queues
Tx Data Not Encrypted
[Logout](#)

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Figure 6-4 Hub TX/RX Modem >Tx>Modulator Queue Common Settings

The following settings impact all Queues at the Hub TX/RX Modulator:

- **Desired Drain Time:** User selectable from 100 – 500ms. This is the target maximum time for a packet placed into a Queue to wait to drain to the WAN. Lower settings can be used to show improvement in jitter while higher settings can be used to increase efficiency
- **Drain Time @ Min ModCods:** Read only status as an indicator to the operator that **IF all queues met their Queue Min Es/No at the same time**, it would take this much time to drain all queues and meet the combined Total Desired CIR. If this number is substantially greater than your “Desired Drain Time” you may want to consider increasing the Min Es/No per Queue, reducing your Desired Queue CIR, or increase the overall capacity of your TX link.
- **Total Desired CIR:** The addition of all Queue(s) Desired Queue CIR plus some modem overhead
- **Total CIR @ Min ModCods:** This is the maximum user throughput of the total system if all sites were running at their lowest MODCOD.
- **Queue Live Time:** The maximum amount of time an Ethernet frame can stay in a queue without being transmitted due to drain congestion. After this time the packet will be discarded.

- Queue Remote ID: A 28 character ID that can be set by the operator as a name for the Queue selected.
- ACM section: (see M7XC_Appendix_E_ACM_Operation).

6.3 Hub TX/RX Modem >Tx>MCC VCM/ACM Configuration

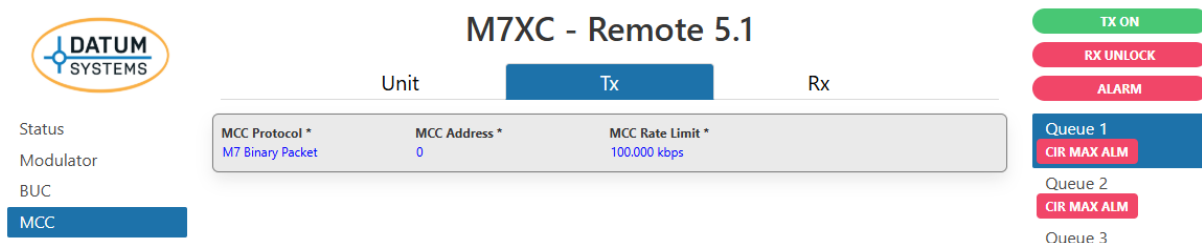


Figure 6-5 TX/RX Modem >Tx>MCC GUI

Set the MCC Address on the >Tx>MCC to a value of “0”. All remotes will use this as their destination address for things like ACM, Remote Es/No etc.

6.4 Hub TX/RX Modem >Rx>Demodulator VCM/ACM Configuration

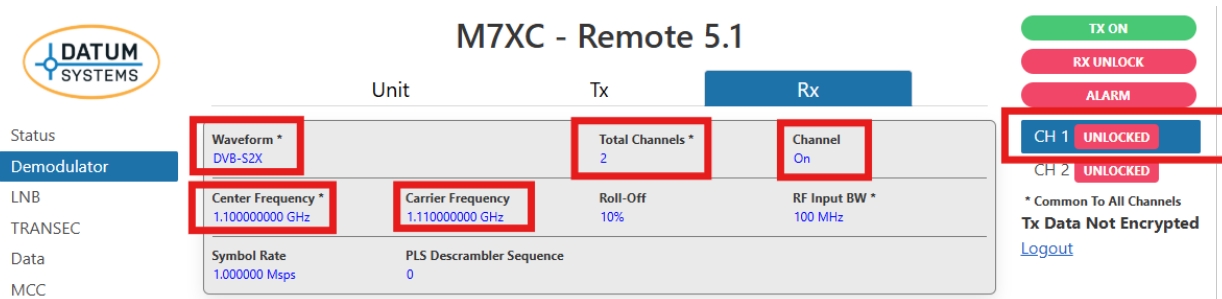


Figure 6-6 TX/RX Modem >Rx>Demodulator GUI

- Waveform*: Set the waveform to match that of the remote site modulator. In this case our Hub Demod Only is receiving multiple DVB-S2X carriers. This can be set to DVB-S2X (as shown) or LDPC-Gen2, or DSSS LDPC-Gen2. DO NOT USE DVB-S2X VCM as this waveform is only received by the remotes, not the Hub.
- Total Channels*: Set this to the total number of channels that the Hub RX-only multi-demod will be using. Maximum demod count depends on the waveform mode.
 - LDPC Gen-2: 1-16 demods
 - DVB-S2X: 1-2 demods
 - DSSS LDPC Gen-2: 1 demod (Total Channels will not appear)
- Channel: Set to On for each channel you are configuring (in the above example we are configuring channel 1 as seen on the right side of the GUI)

- Center Frequency*: This set's the 100MHz center of the multi-demod receiver tuner. All demods configured for this specific modem can only be tuned within this frequency band. Band = center frequency +/- 50MHz.
- Carrier Frequency: This must match the far side remote TX frequency that this demod is landing. In this case channel 1 demod.
- Symbol Rate: Match the symbol rate of the far side remote transmitter symbol rate
- All other parameters configure normally.
- **Repeat steps shown in section 5.4 for ALL CHANNELS (demods) listed on the right side of the GUI.**

6.5 Hub TX/RX Modem >Rx>MCC VCM/ACM Configuration

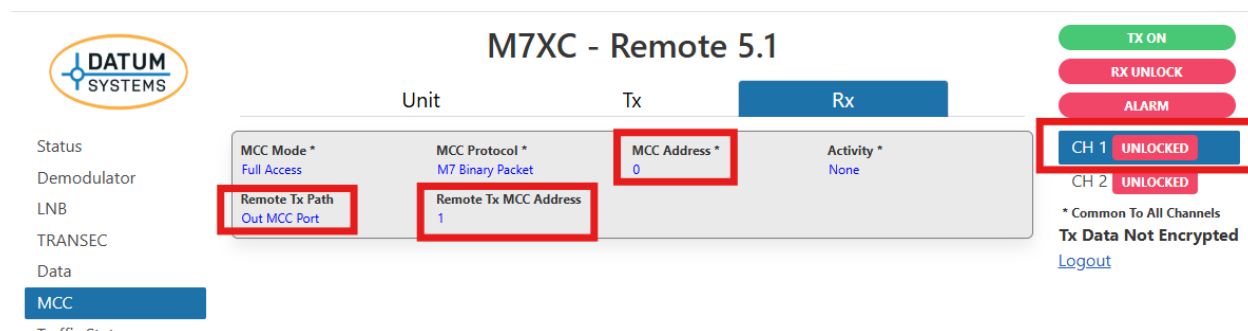


Figure 6-7 Hub TX/RX Modem >Rx>MCC GUI

- MCC Address: Set this to the address of the Hub TX/RX Modem >TX>MCC address.
- Remote Tx Path: For all demods that are in the same physical unit as the modulator (all in the Hub TX/RX Modem) this should be set to "Out MCC Port" this will be different on the Hub RX-ONLY Demod configured later)
- Remote Tx MCC Address: This should be set to match the Tx MCC address of the Remote Modem the demodulator is landing. In our case we are configuring demod #1 landing Remote number 1 with Tx MCC address of 1.
- **Repeat steps shown in section 5.4 for ALL CHANNELS (demods) listed on the right side of the GUI.**

7 Hub RX-Only Demod Configuration (optional if you have Hub RX-Only Demods)

7.1 Hub RX-Only Demod >Unit>WAN Data Port Configuration

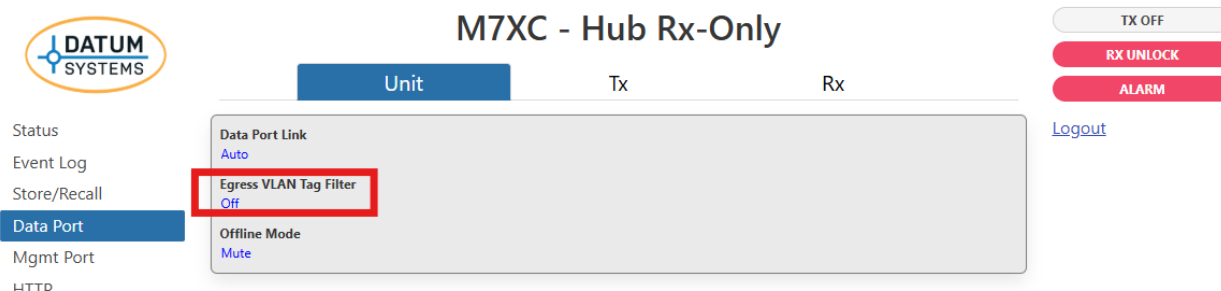


Figure 7-1 Hub RX-Only >Unit>Data Port GUI

- Egress VLAN Tag Filter: Set to Off this will have the effect of sending all traffic to the user data port.

7.2 Hub RX-Only Demod >Tx>Modulator VCM/ACM Configuration

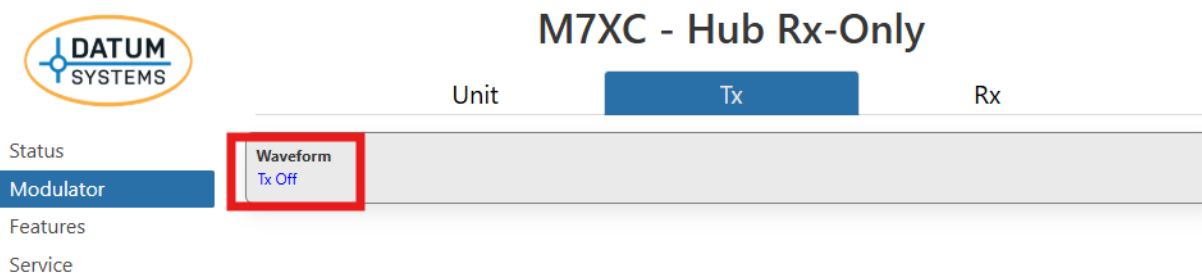


Figure 7-2 Hub Rx-Only >Tx>Modulator GUI

- Waveform: Set to OFF as this is a demod only. All other parameters will be hidden.

7.3 Hub Rx-Only Demod >Tx>MCC VCM/ACM Configuration

- Note the >TX>MCC page will be hidden as there is no modulator configured

7.4 Hub RX-Only Demod >Rx>Demodulator VCM/ACM Configuration

M7XC - Hub Rx-Only

Unit	Tx	Rx
Waveform *	Total Channels *	Channel
DVB-S2X	2	On
Center Frequency *	Carrier Frequency	Roll-Off
1.105000000 GHz	1.100000000 GHz	10%
Symbol Rate	PLS Descrambler Sequence	RF Input BW *
50.000000 Msps	0	100 MHz

CH 1 UNLOCKED
CH 2 UNLOCKED

* Common To All Channels
[Logout](#)

Figure 7-3 HUB RX-ONLY >Rx>Demodulator GUI

- **Waveform*:** Set the waveform to match that of the remote site modulator. In this case our Hub Demod Only is receiving multiple DVB-S2X carriers. This can be set to DVB-S2X (as shown) or LDPC-Gen2, or DSSS LDPC-Gen2. **DO NOT USE DVB-S2X VCM** as this waveform is only received by the remotes, not the Hub.
- **Total Channels*:** Set this to the total number of channels that the Hub RX-only multi-demod will be using. Maximum demod count depends on the waveform mode.
 - LDPC Gen-2: 1-16 demods
 - DVB-S2X: 1-2 demods
 - DSSS LDPC Gen-2: 1 demod (Total Channels will not appear)
- **Channel:** Set to On for each channel you are configuring (in the above example we are configuring channel 1 as seen on the right side of the GUI)
- **Center Frequency*:** This set's the 100MHz center of the multi-demod receiver tuner. All demods configured for this specific modem can only be tuned within this frequency band. Band = center frequency +/- 50MHz.
- **Carrier Frequency:** This must match the far side remote TX frequency that this demod is landing. In this case channel 1 demod.
- **Symbol Rate:** Match the symbol rate of the far side remote transmitter symbol rate
- All other parameters configure normally.
- **Repeat steps shown in section 6.4 for ALL CHANNELS (demods) listed on the right side of the GUI.**

7.5 Hub RX-Only Demod >Rx>MCC VCM/ACM Configuration

The screenshot shows the 'M7XC - Hub Rx-Only' configuration page. On the left is a sidebar with links: Status, Demodulator, LNB, Data, MCC (selected), and Traffic State. The main area has tabs for Unit, Tx, and Rx. Under the Rx tab, there are four fields: 'MCC Mode *' (Full Access), 'MCC Protocol *' (M7 Binary Packet), 'MCC Address *' (0), and 'Activity *' (None). Below these are four red-bordered boxes containing: 'Remote Tx Path' (Out Data Port), 'Tx Hub MAC Address *' (00:19:78:00:00:00), 'Tx Hub Channel ID' (3), and 'Remote Tx MCC Address' (3). On the right, there are buttons for TX OFF, RX UNLOCK, ALARM, and channel status for CH 1 and CH 2 (both UNLOCKED). A 'Logout' link is at the bottom right.

Figure 7-4 Hub RX-Only >Rx>MCC GUI

- **MCC Address*:** Match the MCC address of the Hub TX/RX Modem TX MCC address. In this case 0.
- **Remote Tx Path:** For the Hub RX-Only demodulators this MUST BE SET to “Out Data Port”. This will allow ACM and other special messages to be set to the Hub’s modulator that is performing ACM. When this selection is made you will see an entry for the “Tx Hub MAC Address”
- **Tx Hub MAC Address*:** This is Hub TX/RX Modem’s MAC address it can be found by going to the Hub TX/RX Modem and searching looking at the >Unit>Mgmt Port GUI screen and locating the “MAC Address” field (see below example).

The screenshot shows the 'Mgmt Port' configuration page. The sidebar on the left includes: Status, Event Log, Store/Recall, Data Port, Mgmt Port (selected), HTTP, and SNMP. The main area has tabs for Unit, Tx, and Rx. Under the Unit tab, there are four fields: 'Mgmt Port Link' (Auto), 'DHCP' (Off), 'Address/Mask' (192.168.0.234/24), and 'VLAN ID' (None). Below these are 'Gateway' (Off) and 'MCC Bridge' (Off). At the bottom, a red-bordered box highlights the 'MAC Address' field with the value 00:19:78:00:32:84.

Figure 7-5 Locating Hub TX/RX Modem MAC Address

- **Tx Hub Channel ID:** This must match the far side remote’s “Queue Paired Rx Channel” on the Hub TX/RX Modem >Tx>Modulator section as set in 5.2 for this particular remote. It should also match the “Receive From Tx Queue” of the remote site >Rx>Demodulator as seen in section XXXXX of the remote site that the demod is landing. In this example. The Hub RX-Only demod/channel #1 is landing remote site #3 which is configured process data from Hub TX/RX Modem Queue #3.
- **Remote Tx MCC Address:** Must match the far side remote’s Tx MCC address.

8 Remote Modem Configuration

8.1 Remote Modem >Unit>WAN Data Port Configuration

8.2 Remote Modem >Tx>Modulator VCM/ACM Configuration

8.3 Remote Modem >Tx>MCC VCM/ACM Configuration

8.4 Remote Modem >Rx>Demodulator VCM/ACM Configuration

8.5 Remote Modem >Rx>MCC VCM/ACM Configuration

8.6 Inbound VCM/ACM Demodulator Remote Site

Each remote site in a DVB-S2X-VCM system must configure their demodulator to interface with the Hub outbound modulator. When configuring a remote site demod the following settings are available:

M7XC - Remote 1

Unit Tx **Rx**

Waveform: DVB-S2X VCM

Carrier Frequency	Minimum +/- Acquisition Range	Roll-Off
1.000000000 GHz	30.000 kHz	10%
Symbol Rate	PLS Descrambler Sequence	Receive From Tx Queue
50.0000000 Msps	0	1

TX ON
RX LOCK
NO ALARM

Tx Data Not Encrypted
[Logout](#)

Status
Demodulator
LNB
TRANSEC
MCC
Traffic Stats
Alarms
Events
Features
Service

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Figure 8-1 Remote Modem >Rx>Demodulator Configuration

Goto the >Rx>Demodulator tab.

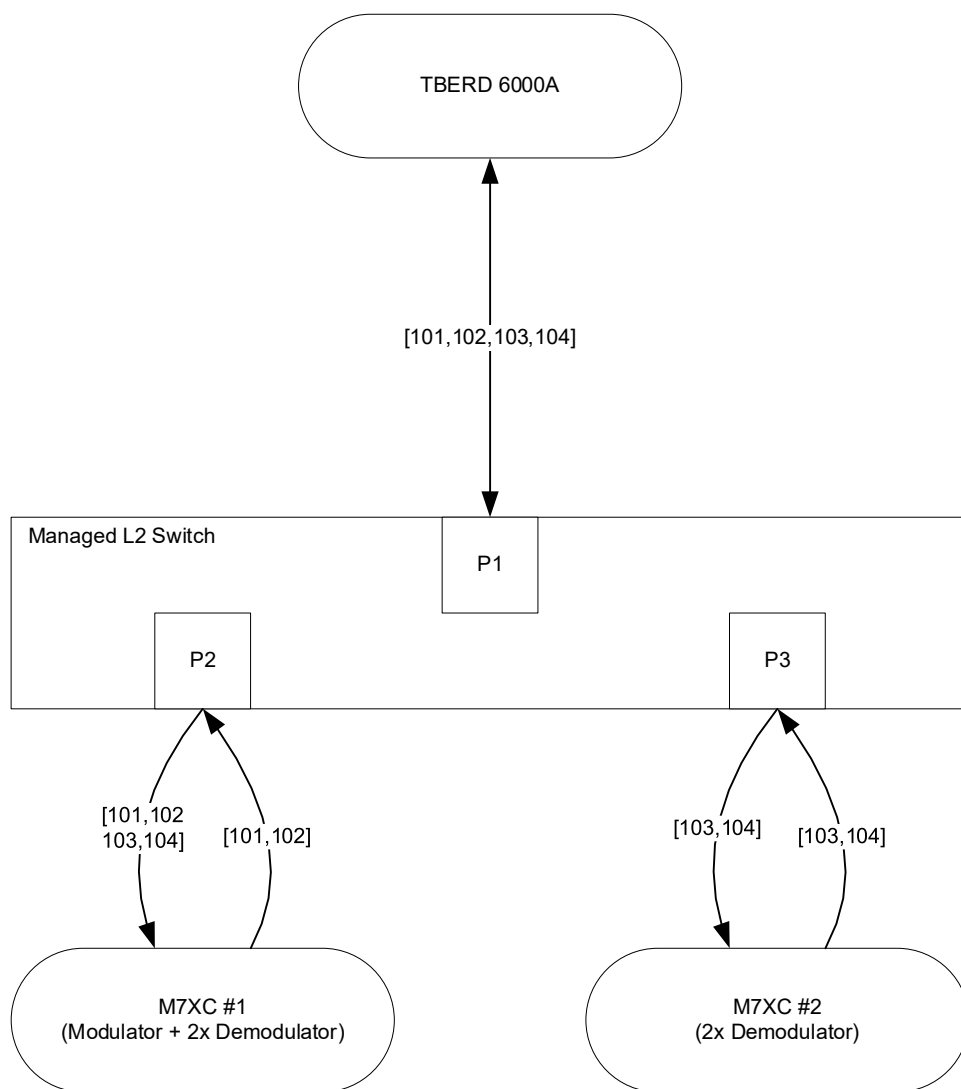
- Demodulator: Be sure this is configured for DVB-S2X VCM for proper operation
- Carrier Frequency: Set frequency to match the far side Hub TX carrier frequency
- Minimum +/- Acquisition Range: Standard operation is to set this to 10% of the hub TX symbol rate up to a maximum of +/- 250KHz. Any carrier that is 2.5Msps or larger use the +/-250KHz setting. Narrower setting for smaller outbound Hub carriers can be used for faster acquisition but going too narrow may cause the carrier acquisition to be too narrow to acquire.
- Roll-Off: Match the Hub TX carrier ROF
- Symbol Rate: Match the Hub TX SR
- PLS Descrambler Sequence: Match the Hub PLS Scrambler Sequence
- Receive From TX Queue: Select the Queue that you want to receive at the site (match from the >Hub>TX>Modulator> Queue Paired Rx Channel).

After this is completed the remote site configured will only receive VLANs associated with the Hub Queue VLAN ID associated with the Queue Paired Rx Channel you have selected (as well as any Broadcast configured Queue).

9 Hub Switch Configuration (3rd party example only)

VCM L2 Managed Switch Setup

This configuration is used for testing the DVB-S2X VCM waveform in a 4 remote system. A diagram of the Hub side of the system is shown below:



The TBERD 6000A is the Hub network. M7XC #1 transmits the outbound shared carrier to all four remotes. M7XC #1 also receives the return channels from remotes #1 and #2. M7XC receives the return channels from remotes #3 and #4.

M7XC#1 maps VLAN 101 to remote #1, VLAN 102 to remote #2, VLAN 103 to remote #3 and VLAN 104 to remote #4.

The traffic between the TBERD and switch port 1 (P1) is a VLAN trunk carrying traffic for all remotes in VLANs 101, 102, 103, 104. The traffic between switch port 2 (P2) and M7XC #1 carries transmit traffic for all remotes in VLANs 101, 102, 103, 104, and carries receive traffic from remote #1 and remote #2 in VLANs 101, 102. The traffic between switch port 3 (P3) and M7XC #2 carries transmit traffic in VLANs 103, 104 which is dropped by M7XC #2, and carries receive traffic from remote #3 and remote #4 in VLANs 103 and 104.

Untagged traffic must go between P2 and P3 to allow ACM Es/N0 packets from P3 to P2 and the ACK for those packets from P3 to P2.

An unmanaged switch cannot be used, and return traffic from VLANs 103 and 104 is learned at P3, and the switch will then send transmit traffic for VLANs 103 and 104 to P3 instead of P2.

The solution is to apply the appropriate VLANs to P1, P2 and P3, and mirror the transmit traffic from P3 to P2. When mirroring the transmit traffic from P3 to P2, P2 should have "Network Traffic" = "Enabled" (Cisco) or "Ingress State" = "Enable" (Planet).

Setup:

1. VCM TRAFFIC

- 1.1. Create VLANs 101, 102, 103 & 104.
- 1.2. Assign VLAN 101, 102, 103 & 104 to P1 (tagged).
- 1.3. Assign VLAN 101 & 102 to P2 (tagged).
- 1.4. Assign VLAN 103 & 104 to P3 (tagged)
- 1.5. Mirror P3 Tx (Egress) to P2 with Traffic=Enabled or Ingress State=Enable.

2. ACM TRAFFIC (optional, this will prevent ACM traffic from hitting the TBERD)

- 2.1. Create VLAN 2
- 2.2. Assign VLAN 2 to P2 (untagged).
- 2.3. Assign VLAN 2 to P3 (untagged).

Attached are two configuration files, one for PLANET IGS-801M, and one for Cisco C1300-8FP-2G

[planet@100-startup-config.cfg](#) (PLANET IGS-801M).

[cisco-p3-p2-mirror-works.txt](#) (Cisco C1300).



cisco-p3-p2-mirror-
works.txt



planet@100-startu
p-config.cfg